

Foundation (Jalapeno)

- Q1.** In a box-and-whisker plot, what does the line inside the box represent?
(A) Median
(B) Mean
(C) Mode
(D) Range
(E) Interquartile Range
- Q2.** What is the primary purpose of a box-and-whisker plot?
(A) To display the mean of a dataset
(B) To show the range of a dataset
(C) To visualize the distribution of a dataset
(D) To compare multiple datasets
(E) To calculate the standard deviation
- Q3.** A scientist records the temperature of a planet in degrees Celsius: $-10, 0, 5, 10, 15, 20, 25$.
(A) 0
(B) 5
(C) 10
(D) 15
(E) 20
- Q4.** The diameter of a black hole is measured in kilometers: 1000, 1200, 1500, 1800, 2000, 2500.
(A) 1200
(B) 1500
(C) 1800
(D) 2000
(E) 2500
- Q5.** What is the interquartile range (IQR) of the dataset
- 2, 4, 6, 8, 10?
(A) 2
(B) 4
(C) 6
(D) 8
(E) 10
- Q6.** The following box-and-whisker plot represents the scores of a physics exam: $Q_1 = 60, Q_3 = 80$.
(A) 20
(B) 30
(C) 40
(D) 50
(E) 60
- Q7.** A box-and-whisker plot shows the following values: $Min = 10, Q_1 = 20, Median = 30, Q_3 = 40, Max = 50$. What is the range of the data?
(A) 20
(B) 30
(C) 40
(D) 50
(E) 60
- Q8.** Which of the following is an outlier in the dataset 1, 2, 3, 4, 5, 100?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 100

Open Ended Questions (FRQ)

- Q9.** Create a box-and-whisker plot to represent the following dataset of planetary distances from the sun in millions of kilometers: 50, 70, 100, 120, 150, 200, 250. Identify the quartiles and interquartile range.
- Q10.** The speed of a spacecraft is recorded in kilometers per hour: 50, 60, 70, 80, 90, 100. Determine the median and interquartile range. Show your calculations and provide a brief explanation of your results.

Chef's Tips

- Tip 1: Remind students to label their diagrams and plots clearly.
- Tip 2: Encourage students to use a ruler to draw their box-and-whisker plots.
- Tip 3: Emphasize the importance of accurately identifying quartiles and the interquartile range.